

Linear Algebra - CS - 3A & 3B

Problem Sheet: Linear Transformation and Their Matrices

Course Facilitator : AH Sheikh

A Linear Transformation and their matrices

1. (**Mechanical**) Define $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by $T(x, y) = (x + 2y, 3x + y)$. Verify that T is linear.
2. (**Conceptual**) Give an example of a function $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ that is not linear. Explain why it fails.
3. (**Mechanical**) Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be defined by $T(x, y, z) = (x - y, y + z)$. Compute $T(1, 2, 3)$ and check additivity for two vectors of your choice.
4. (**Conceptual**) Is there a linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ that is onto? Why or why not?
5. (**Mechanical**) Define $T : \mathbb{R}^4 \rightarrow \mathbb{R}^4$ by $T(x_1, x_2, x_3, x_4) = (x_1, x_3, x_2, x_4)$. Show it is linear and describe geometrically what it does.
6. (**Conceptual**) If $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is linear, what must be true about $T(\mathbf{0})$? Prove it from the definition.
7. (**Mechanical**) Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the projection onto the line $y = x$ along the direction of the y -axis. Find a formula for $T(x, y)$ and verify linearity.
8. (**Conceptual**) Explain why every linear transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ can be represented by a matrix. What are the columns of that matrix?
9. (**Mechanical**) Find a linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ such that $T(1, 0, 0) = (1, 1, 0)$, $T(0, 1, 0) = (1, 0, 1)$, and $T(0, 0, 1) = (0, 1, 1)$. Then compute $T(2, 3, 4)$.
10. (**Conceptual**) Can a linear transformation from $\mathbb{R}^3$ to $\mathbb{R}^4$ be one-to-one? Can it be onto? Justify using dimensions.

5. Matrices of Linear Transformations (10 questions)

1. (**Mechanical**) Find the matrix of $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $T(x, y) = (2x - y, x + 3y)$ relative to the standard basis.
2. (**Conceptual**) What is the matrix of the identity transformation $I : \mathbb{R}^n \rightarrow \mathbb{R}^n$? What about the zero transformation?
3. (**Mechanical**) Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be rotation by 90° around the z -axis (counterclockwise when looking from above). Find the matrix of T relative to the standard basis.
4. (**Conceptual**) If the matrix of $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ relative to the standard basis, what is the matrix relative to the basis $\mathcal{B} = \{(1, 1), (1, -1)\}$?
5. (**Mechanical**) For $T : \mathbb{R}^4 \rightarrow \mathbb{R}^3$ defined by $T(x_1, x_2, x_3, x_4) = (x_1 + x_2, x_2 + x_3, x_3 + x_4)$, find its standard matrix.
6. (**Conceptual**) Explain why different bases give different matrix representations of the same linear transformation. How are these matrices related?
7. (**Mechanical**) Given the matrix $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, describe the linear transformation $T(\mathbf{x}) = A\mathbf{x}$ geometrically. Apply it to the vector $(2, 1)$.
8. (**Conceptual**) If $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ has matrix A , what is the matrix of cT (scalar multiple) and of $T_1 + T_2$ (sum of two transformations)?
9. (**Mechanical**) Find the standard matrix of the linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ that first reflects across the x -axis and then rotates by 45° . Compute $T(1, 0)$ and $T(0, 1)$.
10. (**Conceptual**) Suppose $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ and $S : \mathbb{R}^2 \rightarrow \mathbb{R}^4$ are linear with matrices A (2×3) and B (4×2). What is the matrix of $S \circ T$? What are its dimensions? Prove that matrix multiplication corresponds to composition.